

WRRRA M 0 11 Sequential Calibration of the First Five Inputs

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Abstract

On fixed WRRRA Core 1.0 and MCC 2.3.2, WRRRA_M calibrates its first five physical inputs in the order $G \rightarrow H_0 \rightarrow f_{\text{phi}} \rightarrow f_{\text{c}} \rightarrow \text{electron rest energy}$. Each partial stage reads only supplied anchors and reports available outputs and remaining coefficients. The 0.10 energy-pressure calculation and nine physical cases are retained. Address-exponent and effective-admission reparameterization explicitly discloses additional arithmetic inputs. Adopting electron mode 23 and zero intercept gives a mass unit of 22,217.345682 eV. Two response shapes fitting the same anchors demonstrate reproduction together with residual freedoms.

Verification inputs

The complete 0.10 input is frozen by hash. G and electron energy use NIST 2022 CODATA; $H_0=67.4$ is the Planck 2018 base-LambdaCDM calibration benchmark. CODATA standard uncertainties are 1.5×10^{-15} SI for G and 0.00016 eV for electron energy. The adopted H_0 interval of $\pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is the Planck base-LambdaCDM 68% confidence interval. Physical fractions 4.93% and 26.5% are inherited rounded calibrations without a covariance assigned here. The first five candidate-sheet entries and the MCC electron-mode relation are recorded as provenance.

Input	Adopted value	Unit / role
G	6.6743000e-11	$\text{m}^3 \text{ kg}^{-1} \text{ s}^{-2}$
H_0	67.4	$\text{km s}^{-1} \text{ Mpc}^{-1}$
f_{phi}	0.0493	energy fraction
f_{c}	0.265	energy fraction
$m_{\text{e}} c^2$	510998.95069	eV

$c=299792458 \text{ m/s}$ is fixed. $h=6.62607015 \times 10^{-34} \text{ J s}$ and $1 \text{ eV}=1.602176634 \times 10^{-19} \text{ J}$ are exact SI references. The parsec conversion, local masses/radii/response function and conditional $\Phi=\Psi$ lensing are inherited. Address cutoff, four zeta heights, $K=8$, phase increment 0.1, response coefficients λ_s , volume exponents and the electron-mode integer are constitutive inputs; the five numerical anchors do not derive them.

WRRRA-specific transformation

The partial calculator adds one anchor at a time. G fixes the SI action normalization and curvature response. Exact h and c additionally give Planck length/time as unit references. Replacing all future anchors with NaN must leave the corresponding earlier-stage outputs unchanged.

$$C_G = \frac{c^4}{16\pi G}, B_G = \frac{8\pi G}{c^4}, \ell_P = \sqrt{\frac{\hbar G}{c^3}}, t_P = \ell_P / c. \quad (1)$$

After converting H0 to inverse seconds, the critical energy density is calculated. c/H0 and 1/H0 are Hubble length/time references, not assignments of the total cosmic size or age.

$$H_0 = \frac{H_{0, km/s/Mpc} 10^3}{10^6 pc_m}, u_{crit} = \frac{3 H_0^2 c^2}{8 \pi G}. \quad (2)$$

The phenotype fraction gives the total hidden load. Adding the clustering fraction f_c separates D from background R. eta_load is the homogeneous-carrier reference coefficient of 0.6, distinct from the address-sector eta_s coefficients of 0.10.

$$f_h = 1 - f_\varphi, f_b = 1 - f_\varphi - f_c, u_h = f_h u_{crit}, \eta_{load} = u_h / 2. \quad (3)$$

$$Q_h \equiv \zeta k_h^2 = \frac{16 \pi G u_h}{c^4}, L_* = Q_h^{-1/2}, L_0 = L_* \sqrt{W_0}. \quad (4)$$

The identifiable quantity is weighted twist Q_h. Stiffness zeta and twist kappa_h are not separately determined. L_star is a calculated length coefficient; L0 remains undetermined without W0. The 95.07% hidden share alone does not fix cosmic length.

Step	New input	Newly available outputs	Remaining anchors / choices
1	G	action coefficient; SI response	H0, f_phi, f_c, E_e
2	H0	H0 in s ⁻¹ ; critical density	f_phi, f_c, E_e
3	f_phi	hidden density; weighted twist	f_c, E_e
4	f_c	sector energies; a_T, P, q	E_e
5	E_e	mass unit; electron units	structural freedoms remain

The 0.10 address effects e_s(n), normalized weights w_n and positive responses g_s(n) are retained. Each eta_s is fitted once on the reference address state and uniform carrier, then held fixed for address/state controls. Carrier operators are A_phi=I, A_D=Kc/2 and A_R=Kb/2, each with unit uniform reference expectation.

$$\mu_s = \sum_n w_n e_s(n) g_s(n), \eta_s = \frac{f_s u_{crit}}{\mu_s}, \hat{E}(a) = V_0 \sum_s \eta_s \mu_s a^{\nu_s} A_s. \quad (5)$$

Pressure is the volume derivative of the same energy function. Retaining nu_phi=nu_D=0 and nu_R=3 gives pressureless D/phenotype and P_R=-u_R. Nonphenotype status does not delete energy or gravity.

$$V = V_0 a^3, E_s = \text{Tr}(\rho \hat{E}_s), P_s = -\partial E_s / \partial V = -\nu_s u_s / 3. \quad (6)$$

$$a_T = c H_0 \sqrt{f_c / 8}, P_0 = -f_b u_{crit}, q_0 = (1 - 3 f_b) / 2. \quad (7)$$

Electron mode n_e=23 is adopted in the zero-intercept MCC mass branch. The electron anchor fixes its common energy unit. Reported ±1, ±2, ±3 and ±23 are generic mode controls, not identifications of other particle masses. Mode sign has the same rest energy; charge remains separately inherited from the filter ledger. Mode number and zero intercept are additional constitutive choices.

$$E_n = \sqrt{E_{off}^2 + n^2 \mu_E^2}, E_{off} = 0, \mu_E = E_e / 23, E_e = m_e c^2. \quad (8)$$

The same total energy operator is expressed in electron-energy units. Multiplying back by E_e restores the original joule ledger. Electron energy is not separately added to cosmic density. Physical boundary conditions, the microscopic clock and general spectra remain for 0.12.

$$\tilde{E}_e = \hat{E} / E_e, \hat{E} = E_e \tilde{E}_e, m_e = E_e / c^2. \quad (9)$$

Re-expression of address alpha and beta

Here α_{addr} is the address power-law exponent and β_{eff} is the effective admission rate of odd composites. These differ from electromagnetic fine-structure α_{EM} . The two independent 0.9 arithmetic targets $D=0.268$ and $\phi=0.05$ remain disclosed additional inputs; physical energy targets 0.265 and 0.0493 do not replace them. C_{even} excludes the prime address 2.

$$w_n(\alpha_{addr}) = \frac{n^{-\alpha_{addr}}}{\sum_{j=2}^N j^{-\alpha_{addr}}}, \sum_{n \in C_{even}} w_n = 0.268. \quad (10)$$

$$\sum_{n \in C_{odd}} w_n b_n(h) = 0.05, \beta_{eff} = \frac{0.05}{\sum_{n \in C_{odd}} w_n}. \quad (11)$$

Resolving the fixed phase/filter gives $\alpha_{addr}=1.899687695055$, $h=1.447673170353$ and $\beta_{eff}=0.865457012414$. α_{addr} and h fit the two arithmetic targets; β_{eff} is then derived as their re-expression. A third independent beta input is unnecessary. The five physical anchors alone do not determine arithmetic targets or α_{EM} .

Calculated outputs

Quantity	Calculated value
ucrit J m ⁻³	7.668947767822e-10
u_hidden J m ⁻³	7.290868642868e-10
eta_load J m ⁻³	3.645434321434e-10
zeta kappa_h ² m ⁻²	3.028112744666e-52
Length coefficient m	5.746639841098e+25
a_T m s ⁻²	1.191812669106e-10
P Pa	-5.258597484395e-10
q0	-5.285500000000e-01
mu_E eV	2.221734568217e+04

The reference retains $q0=-0.528550000$, local rotation speed 207.510905127 km/s and conditional lens deflection 0.535586511 arcsec. The mass unit is 22217.345682174 eV. These are reproduction of validated benchmarks

and internal calculations under disclosed inputs. A new independent prediction is not a prerequisite for completion.

Residual freedoms are shown by calculated controls. Rescaling zeta and kappa_h as below preserves Q_h. Separately fitting eta_R at lambda_R=0 and 1 reproduces the same five targets and reference q, yet changing K to 4 after calibration gives different results. Fitting reference products eta_s mu_s and determining the full response shape are separate achievements.

$$\zeta \mapsto r \zeta, \kappa_h \mapsto \kappa_h / \sqrt{r}, \zeta \kappa_h^2 \mapsto Q_h. \quad (12)$$

lambda_R	Reference q	K=4 q after fitting
0.0	-0.528550000	-0.547819855
1.0	-0.528550000	-0.548643149

Independence of the five anchors is checked using the logarithmic sensitivity of $(C_G, u_{crit}, u_h, u_c, \mu_E)$. The numerical rank is five conditional on frozen constitutive choices. This is local identification of the five calibration anchors, not identification of every model coefficient.

$$\frac{\partial \log(C_G, u_{crit}, u_h, u_c, \mu_E)}{\partial \log(G, H_0, f_\phi, f_c, E_e)} = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 & 0 \\ -1 & 2 & -f_\phi/(1-f_\phi) & 0 & 0 \\ -1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \quad (13)$$

Ten runs changing each anchor by $\pm 1\%$ are construction-sensitivity controls, not observational confidence intervals. Changed physical targets are explicitly recalibrated; address controls hold SI coefficients fixed. Electron energy changes the mass unit but not macroscopic density, q or rotation speed. Critical density varies inversely with G and quadratically with H0.

The fixed-reference homogeneous expansion uses the same pressureless/background energy relation. Noncommuting-carrier internal exchange at fixed volume sums to zero and preserves state positivity/normalization. The required conversion work and its environmental account retain the 0.10 conservation checks.

Revision r1 retains the clock verdict established in 0.12. Fixed-volume state evolution connects to SI phase through $\xi = E_{\text{star}} \tau_{\text{phys}} / \hbar$. The slow nonuniform expansion schedule of 0.10 differs from that SI phase, so completion of sequential calibration does not imply completion of covariant nonuniform dynamics. The uniform I/N state is stationary under unitary updates and retains the inputs and outputs of the expansion relation below.

$$\frac{H|a|^2}{H_0^2} = (f_\phi + f_c) a^{-3} + f_b, \frac{du}{d \log a} + 3(u + P) = 0. \quad (14)$$

Falsifiers and reproduction

The connection must be revised if an earlier stage reads future targets, input order changes, energy-pressure differentiation or inherited benchmark reproduction fails, electron energy is duplicated, or unidentified W0, zeta or alpha_EM is used as determined. This release executes 28 checks plus 32 inherited 0.10 checks, including zero/sign modes, five-point pressure differentiation, log sensitivities, explicit degeneracy controls, conservation and invalid-input rejection.

```
python -m pip install -r calculations/wrra_m_0_11/requirements.txt
python calculations/wrra_m_0_11/run_release.py
```

Parameters.json contains inputs, units, provenance and constitutive choices. Sequential calibration, physical cases, mass modes and input responses are supplied as JSON/CSV. A clean copy with all captured results removed compares regenerated outputs and bilingual sources byte for byte. Word/PDF are final publications built from those sources, with equations/tables checked against execution and every rendered page inspected.

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Input SHA256 67de28abf0741c406178a2fe4ce07962e69bce827c3196b78e887da3bd0dab13

Completion and subsequent scope

Version 0.11 completes staged calibration of the five disclosed anchors and reporting of residual freedoms. Their origins, a unique universe solution and calibration-free constants are not completion requirements. Version 0.12 treats clocks/boundaries/spectra, 0.13 physical quantization/post-observation states/records, and later stages uncertainty, stress, curvature, size and neutrino connections. Unmeasured outputs produced after fixing the model and calibration are recorded as WRRRA predictions with their inputs and falsifiers.

References

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